MIME 497 Capstone Design Proposal

SUMO BOT

Project Number: ME-615B

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1 Project Scope and Context

1.1 Project Overview and Background

This capstone project involves designing and building a 3 kg remote-controlled sumo robot for competition at the Engineering Expo in June 2026. The robots must follow Robogames 3 kg RC sumo regulations and compete within a 154 cm diameter circular ring called a dohyo. Competition scoring follows Unified Sumo Robot rules where the first team to win two rounds receives two Yuhkoh points and wins the match.

Analysis of existing sumo robot designs reveals common successful strategies including frontal ramps for pushing opponents, treads for increased traction, sharp angle ramps for wedging underneath opponents, and low centers of gravity for stability. Control methods typically involve either wireless RC transmitters or autonomous sensors. Our sponsor relayed this project will be using a remote control system.

1.2 Customer Requirements

Through discussions with our project sponsor and analysis of competition designs, we compiled this list of customer requirements for our House of Quality (HOQ).

CR	Description	Weight	Engineering Specification & Target
1	Complex Geometry	15	Dimensions: non-cube geometry
2	RC Controlled (NO Sensor)	10	Control input sources: RC controlled
3	Battery Longevity	15	Runtime: 8hr total competition time including recharge cycles
4	Easy Accessible Interior	5	Access: hole > 5 in diameter
5	Must be ready to compete in Week 7 of second term	10	Timeline: Ready by end of May
6	Conforms with maximum size and weight requirements	20	Weight: 3 kg, -0.5 kg tolerance; Footprint: 20 cm × 20 cm, -5 cm tolerance
7	Offensive/Defensive Feature	5	Mechanism effectiveness: Force > Weight of opponent bot
8	Durability	10	Structural strength: No permanent deflection through competition
9	Stability	10	Low center of mass: Center of mass < 25% of total height

Table 1: Customer Requirements with Weights and Engineering Specifications

1.3 Constraints and Project Resources

The project operates under a $1,000 budget with potential sponsor supplementation and a two-term timeline requiring design completion in MIME 497 and manufacturing completion by Week 7 of MIME 498 (Spring Term). All designs must follow Robogames 3 kg RC sumo guidelines. The team has access to I-Labs, first-floor Rogers materials and tools, and Merriweather 3D printing facilities.

Current estimated budget breakdown. Used 70%, 30% still avalible. Motors ~15%, Wheels ~10%, Battery ~15%, Electronics (Servos, UBEC, Wiring) ~20%, Hardware ~10%

1.4 Stakeholders

The project is sponsored by the OSU School of MIME Capstone Program under Dr. Oman's leadership, with an expectation of a fully functional robot demonstrating efficient engineering practices for competition at Engineering Expo in June 2026. Dr. Oman serves as technical advisor, providing guidance and progress evaluation. The design team consists of Matthew

Brady, Jonah Stuart, and Bella Sifuentez. Personal connections with two other competing teams enable practice matches for performance improvement. Another important stakeholder group includes youth observers at Engineering Expo, as inspiring younger generations to pursue engineering fields through hands-on demonstrations supports continued innovation.

2 Design Process

Our team followed a traditional product development process consisting of research, design, prototyping, and manufacturing phases. This approach allows brainstorming of many design alternatives while maintaining flexibility to iterate based on testing results. The research phase involved analyzing previous sumo robot designs to identify successful strategies including frontal ramps, tread-based traction, sharp-angle wedging geometry, low centers of gravity for stability, and RC control reliability versus autonomous sensors.

To satisfy the requirement for at least one defensive or offensive mechanism beyond a basic ramp, the team brainstormed several supplemental features. Concepts included an aluminum bar attached to a servo for swiping opponents to the side, various lifting mechanisms inspired by battle bots, and alternative ramp geometries for improved wedging capability. The team evaluated these alternatives based on their effectiveness at neutralizing opponent resistance, manufacturability within the project timeline, and compliance with competition rules regarding push-based competition without flipping.

Our project team has been using a working model CAD for the chassis, ramp and a full assembly. Based on the teamates choosen software we convert these models between Solidworks and Simens Nx.

3 Individual Proposed Design Solution

3.1 Solution Overview

The personal design proposal was based around being as innovative as we could without breaking any rules. After discussion with Dr. Oman, it was noted that there was a large grey area in the unified sumo rules that could be exploited to give an advantage. After running through the rules, there was one thing that stood out that could possibly be in the grey area and give the team a slight advantage. The rules are very specific about flipping, but there is no stated documentation regarding lifting the opposing bot. After discovering this, the team did a lot of research on how we could design a mechanism that could efficiently lift the opposing bot and carry it out of the ring.

The biggest benefit of this mechanism is that it would be able to render the opposing bot's pushing force null. During my research on battle bots, I discovered a battle bot called Overhaul. This bot has two separate motors controlling a bottom arm and a claw arm that can work independently and simultaneously. To make this suitable for our sumo robot, I proposed we make a very tight-to-the-ground angle ramp that would act as the bottom arm, and two large outward claw arms that act on another motor. This would allow us to drive under an opposing bot, activate the top arms to clamp, then activate the bottom and top arms to move up simultaneously so that we can lift the opposing bot off the ground. This would take away any ability to resist using friction and resist using motor power.

3.2 Visual Representation

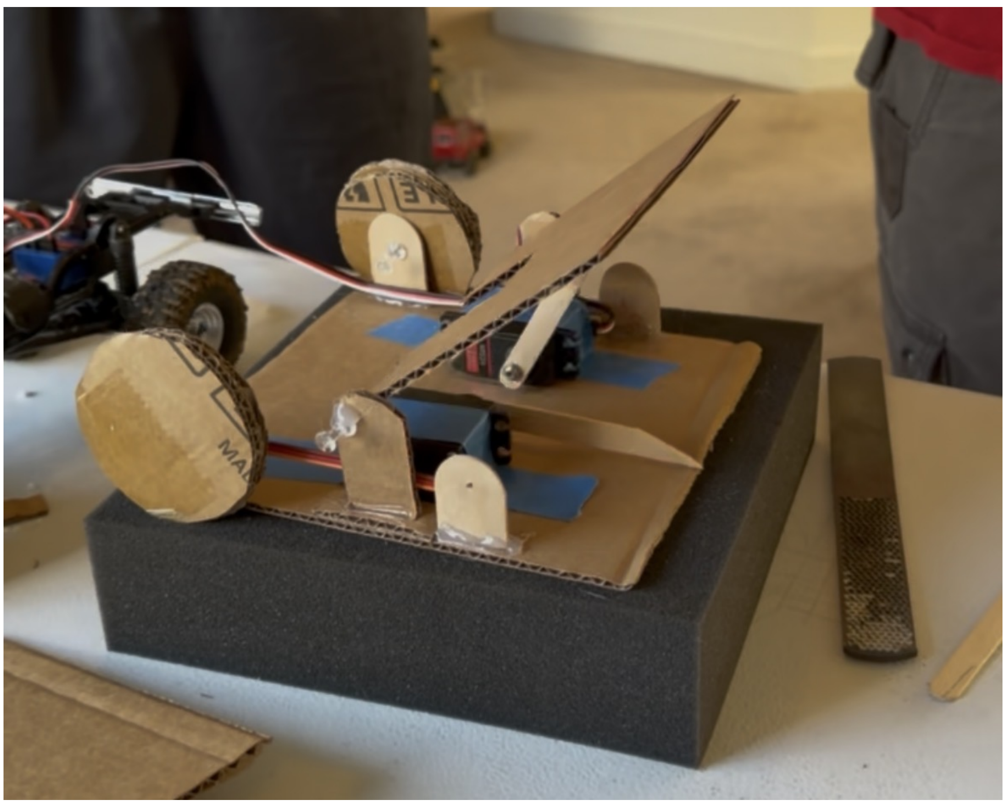

Figure 1: Proposed lifting mechanism design showing low-profile ramp, claw arms, and lifting assembly inspired by Overhaul battle bot

(Fig. 1) shows the three-stage operation of the lifting mechanism. In the initial position, the low-angle ramp sits flush with the ground while the claw arms remain in their open, outward-facing configuration. As the robot drives forward and wedges beneath an opponent, the proximity triggers activation of the claw servos, which rotate the arms inward to secure the opponent between the upper claws and lower ramp. Finally, the lifting motor engages to raise the entire assembly vertically, elevating the captured opponent off the dohyo surface and eliminating any traction advantage.

3.3 Advantages of This Solution

This lifting mechanism design offers several significant advantages when evaluated against the customer requirements. The most substantial benefit addresses the offensive and defensive features by introducing a fundamentally different approach to neutralizing opponent capabilities. Rather than relying solely on pushing force, the lifting mechanism eliminates the opponent's ability to resist by removing physical contact with the dohyo surface. Once lifted, the opponent cannot generate friction-based resistance or utilize motors effectively. This represents a significant tactical advantage exceeding traditional pushing-based strategies.

Regarding complex geometry, the design clearly satisfies the requirement through its integrated lifting assembly and claw arm configuration. The asymmetric arrangement of components and the three-motor system create a distinctive profile differentiating our robot from standard box-based designs. The footprint remains within 20 cm by 20 cm as the claw arms rotate upward before competition.

3.4 Disadvantages and Limitations

The biggest design constraint identified from prototyping is the space needed for all the components. This was tested by cutting out foam models of the parts to scale and trying to fit them into the working prototype.

While the lifting mechanism theoretically renders opponent resistance null by eliminating ground contact, getting the opposing bot into the vice might be difficult due to the opposing bot moving around and possibly its mechanisms.

4 Team Selected Design Solution

Team Sumo Blue had chosen to use a design very similar to what I proposed with the clamp and lift system. That being said, it is not exactly what was originally proposed. The best part about group design work is having even more brains to help increase the design potential to make a good idea even better. Since then, our team has altered the idea of just the claw clamp from the center to raising it to a higher pivot point to both increase the internal space for components and to get a better angle for the top claw arm.

4.1 Visual Representation

Figure 2 is the low-fidelity model of our Version 1 concept. This displays the lower pivot point for our top arms.

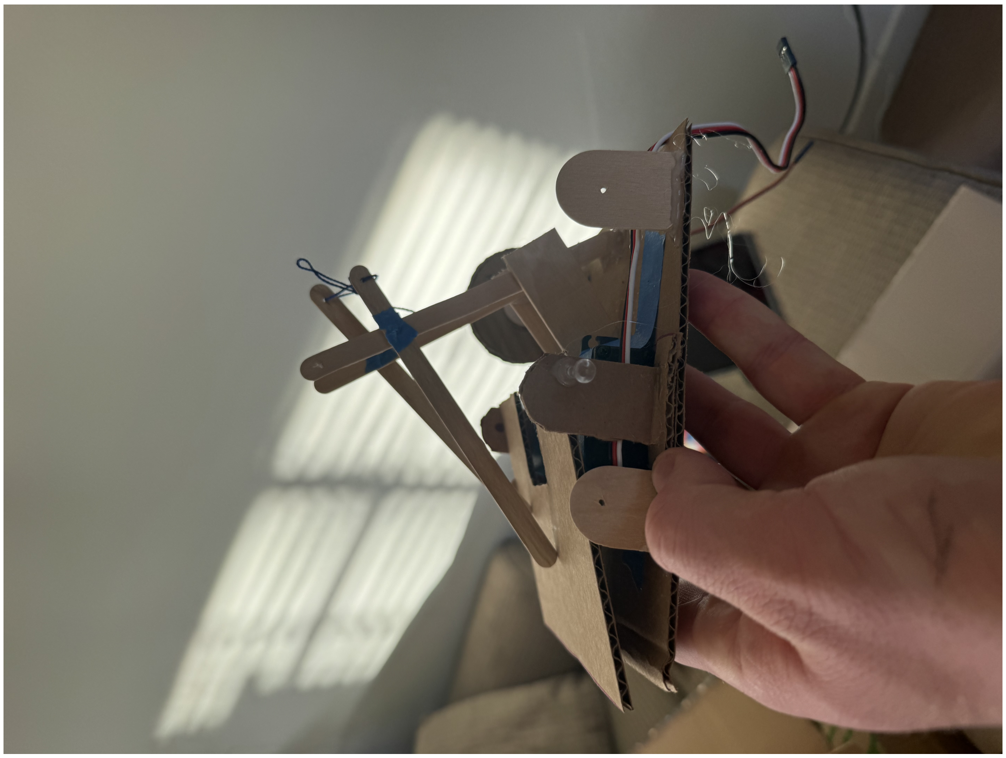

Figure 2: Low-fidelity prototype of Version 1 design showing pivot point configuration

5 Conclusion

At this point in the project, our design choices have been honed down to a very specific set of ideas and need to be quantified by the real parts that will be used to construct the bot. As our project team continue to complete these designs, collaboration will continue to create the best design possible.

Next steps through this project are to continue attacking the design of each subsystem and integrating this into the bots full assembly. The current sub assembly to work on is the top claw servo holder and pivot joints. The ramp also requires attention to ensure that the bottom lifting servos will be able to effectively slide the ramp up.

After those sub assembly have been designed our team will continue with testing and ensure that our bot is able to preform all operations to their full extent.

References

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